

Minh Le Bui

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EDUCATION

University of Wisconsin - Madison

Bachelor of Science, Computer Science & Data Science (GPA: 4.0/4.0)

May. 2028

Madison, WI

- **Awards & Leadership:** Dean's List - All Semesters; Vice President - Vietnamese International Student Association
- **Recognition:** ICPC North Central North America (NCNA) Regional 2025 – Placed 14th out of 86 teams
- **Courses:** Data Structures and Algorithms (A), OOP (A), Distributed System (A), Computer Architecture (A), Data Science Programming (A), Operating System (A), Data Management (A), Statistical Modeling (A), Machine Learning, Database

TECHNICAL SKILLS

Specialization: Machine Learning & LLMs, Data Science, Data & Backend Systems, Cloud Infrastructure

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML, CSS, R

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch

Frameworks & Libraries: PySpark, PyTorch, LightGBM, XGBoost, Transformers, LangChain, FastAPI, Flask, SQLAlchemy, Pandas

Cloud & Tools: AWS, Firebase, Git, GitHub Actions, Docker, Linux, Conda, VMware Fusion, Tableau

EXPERIENCE

CMC AI

Jun. 2026 – Aug. 2026

AI Intern | Transformers, PaddleOCR-VL 1.6, PP-DocLayoutV3, Data Augmentation, XGBoost

Hanoi, Vietnam

- Built a failure taxonomy from 2,000+ **PaddleOCR-VL 1.6** outputs, separating data-driven recognition errors (poor handwriting, faded scans) from post-processing defects (layout breaks, formula formatting), guiding targeted training data collection.
- Designed a data-curation pipeline with confidence-, document-type-, and signature-aware sampling, outperforming the 80K-image baseline (**0.485 → 0.497 mAP**) with only 23.5K images (**3.4× fewer**); checkpoint adopted as the DLA training baseline.
- Developed a **LayoutReader-XGBoost** labeling model to generate reading-order annotations from pre-detected layout boxes, raising exact-match accuracy from **51.3% to 78.9%** and successor F1 from **81.7% to 92.3%** over rule-based labeling on Chunkr.
- Fine-tuned **PP-DocLayoutV3 (RT-DETR)** for legal-document layout detection and reading order, raising COCO mAP from **0.419 to 0.497 (+18.6%)** over SOTA; improved 8/11 classes by up to **+0.18** and added a new signature class at **0.38 AP**.

Thayer School of Engineering at Dartmouth – Advisor: Prof. Eugene Santos

Sep. 2025 – Present

Research Assistant | Llama-3.2-3B, LLM Fine-Tuning, Benchmarking, LangChain

Remote

- Fine-tuned **Llama-3.2-3B** (LoRA/QLoRA/full) to study hallucination, reducing cross-paraphrase answer variance $\sim 3\times$ (**49%→17%**), achieving **100% consistency** on held-out facts; LoRA/QLoRA matched full fine-tuning at far fewer parameters.
- Compared Llama and Mistral on **520 AdvBench** prompts, identifying an **18.3%** refusal gap using refined response classification.
- Developing **Pair-H**, an iterative adversarial question-rewriting pipeline with an **evidence-grounded judge** to induce and detect unsupported or contradicted LLM claims.

VNPT-Media

Jun. 2025 – Aug. 2025

Machine Learning Engineer Intern | XGBoost, Docker, FastAPI, PySpark

Hanoi, Vietnam

- Implemented a **PySpark** feature engineering pipeline, processing **2M+** monthly user records to facilitate monthly retraining and ensure long-term model stability.
- Developed and deployed an **XGBoost model** to predict return likelihood of one-way locked users, achieving a final F1 score of **0.82 (14% improvement)** over the 0.72 Logistic Regression baseline; deployed internally via **Docker** and **FastAPI**.
- Performed systematic experimentation with feature selection and Bayesian **hyperparameter tuning**, leading to **+5%** in predictive performance; leveraged SHAP (Explainable AI) to explain model outputs and support non-technical stakeholders.

Biokind Analytics

Sep. 2024 – May. 2025

Data Engineer & Data Analyst | Pandas, GeoPandas, Matplotlib, Tableau

Madison, WI

- Built automated pipeline to ingest and clean data from 15+ counties via FIPS codes, reducing manual processing time by **40%**.
- Processed and analyzed **12GB+** of 6-year historical data using **Pandas**, directly informing partner bank's donation strategy.
- Developed interactive **Tableau** and **GeoPandas** dashboards for stakeholders to plan upcoming donation campaigns.

TECHNICAL PROJECTS

Crypto Market Anomaly Pipeline | AWS, Terraform, PySpark, SQL

Aug. 2026 – Present

- Built a serverless medallion pipeline (S3, Lambda, Glue/PySpark, Athena, Terraform) turning a live crypto feed into two SQL-queryable tables: clean minute bars and a ranked table of unusual price and volume moves.
- Ingested 1-minute bars for 17 pairs every 5 minutes; dedup on (symbol, open_time) and time-based Spark windows delivered **zero duplicates, zero OHLC violations, and zero gaps** across 16 days.
- Shipped the anomaly layer as per-symbol rolling z-scores (60-bar window, log1p volume, variance guard) flagging **1.33%** of scored bars, with each event carrying its baseline and detection parameters for reproducibility.
- Building an Athena-backed quality gate (8 checks, two severities) that blocks bad data from reaching the analytics layer, plus Step Functions orchestration, backed by **55 mutation-verified PySpark tests**.

AWARDS & ACHIEVEMENTS

Vietnam Science and Engineering Fair (ViSEF) – 4th prize (System Software)